

SQL QUERY AND ANSWERS

Q1. What is the total revenue generated by male vs. female customers?

```
select gender, SUM(purchase_amount) as revenue
```

```
from customer
```

```
group by gender
```

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	gender text	revenue numeric		
1	Female	75191		
2	Male	157890		

Q2. Which customers used a discount but still spent more than the average purchase amount?

```
select customer_id, purchase_amount
```

```
from customer
```

```
where discount_applied = 'Yes' and purchase_amount >= (select AVG(purchase_amount) from customer)
```

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	customer_id bigint	purchase_amount bigint		
1	2	64		
2	3	73		
3	4	90		
4	7	85		
5	9	97		

Q3. Which are the top 5 products with the highest average review rating?

```
select item_purchased, round(avg(review_rating::numeric),2) as "Average Product Rating"
from customer
group by item_purchased
order by avg(review_rating) desc
limit 5
```

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	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Q4. Compare the average Purchase Amounts between Standard and Express Shipping.

```
select shipping_type,
ROUND(AVG(purchase_amount),2)
from customer
where shipping_type in ('Standard','Express')
group by shipping_type;
```

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

```
SELECT subscription_status,
```

```

COUNT(customer_id) AS total_customers,
ROUND(AVG(purchase_amount),2) AS avg_spend,
ROUND(SUM(purchase_amount),2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue,avg_spend DESC;

```

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	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Q6. Which 5 products have the highest percentage of purchases with discounts applied?

```

SELECT item_purchased,
ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)/COUNT(*),2) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;

```

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	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Q7. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

```
with customer_type as (  
SELECT customer_id, previous_purchases,  
CASE  
WHEN previous_purchases = 1 THEN 'New'  
WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'  
ELSE 'Loyal'  
END AS customer_segment  
FROM customer)
```

```
select customer_segment, count(*) AS "Number of Customers"  
from customer_type  
group by customer_segment;
```

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Q8. What are the top 3 most purchased products within each category?

```
WITH item_counts AS (SELECT category, item_purchased,  
COUNT(customer_id) AS total_orders,  
ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS item_rank  
FROM customer
```

```
GROUP BY category, item_purchased
```

```
)
```

```
SELECT item_rank, category, item_purchased, total_orders
```

```
FROM item_counts
```

```
WHERE item_rank <=3;
```

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	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171

Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

```
SELECT subscription_status, COUNT(customer_id) AS repeat_buyers
```

```
FROM customer
```

```
WHERE previous_purchases > 5
```

```
GROUP BY subscription_status;
```

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Q10. What is the revenue contribution of each age group?

```
SELECT age_group, SUM(purchase_amount) AS total_revenue
```

```
FROM customer
```

```
GROUP BY age_group
```

ORDER BY total_revenue desc;

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	subscription_status text	repeat_buyers bigint	
1	No	2518	
2	Yes	958	